Variable Inductor Design Report

Core: **E 30/15/7 TDK** | Generated: 2026-03-24 10:44

Electrical Specifications

Parameter	Symbol	Value	Unit
Main winding nominal inductance (I_bias=0)	L_main_nom	60.0	uH
Main winding minimum inductance (I_bias_max)	L_main_min	6.0	uH
Main winding nominal inductance (achieved)	L_main_nom*	60.00	uH
Main winding minimum inductance (achieved)	L_main_min*	7.37	uH
Main winding frequency	f_sw	100.0	kHz
Main winding RMS current	I_main_rms	1.3	A
Main winding peak current	I_main_pk	3.4	A
Main winding max flux density	B_main_max	0.35	T
Bias winding max DC current	I_bias_max	0.5	A

Core Material

Material: **N87 (TDK)** | TDK EPCOS N87, MnZn, ~100 kHz

Parameter	Symbol	Value	Unit
Initial permeability	mu_i	2200	-
BH model k1	k1	0.062	H/m
BH model k2	k2	42.995	T^-2
BH model k3	k3	302.904	H/m

Core Geometry

Core: **E 30/15/7 TDK** | Set of E + I cores

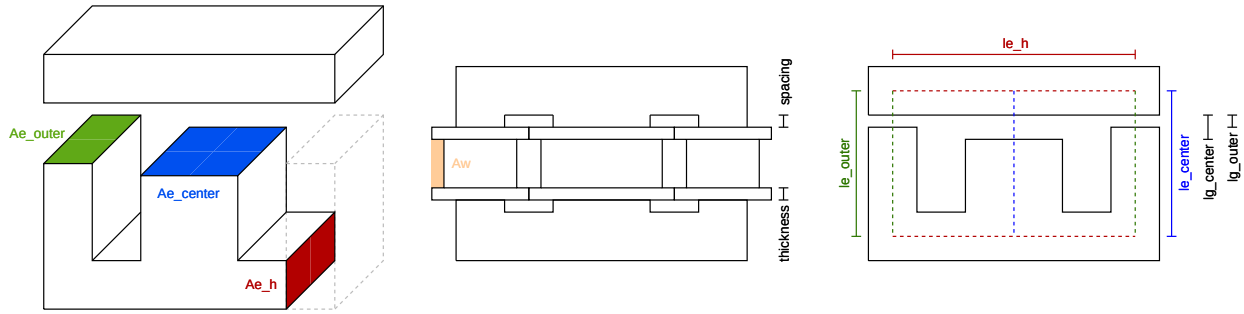
Parameter	Symbol	Value	Unit
Centre-leg cross-section	Ae_center	52.56	mm2
Outer-leg cross-section	Ae_outer	38.32	mm2
Horizontal cross-section	Ae_h	40.15	mm2
Centre-leg path	le_center	24.90	mm
Outer-leg path	le_outer	24.90	mm
Horizontal path	le_h	24.75	mm
Coil former width	CF_W	1.95	mm
Coil former length	CF_L	7.70	mm
Window area	Aw	15.01	mm2
AeAw required	AeAw_req	214.39	mm4
AeAw available	AeAw_avail	789.19	mm4

Air Gaps

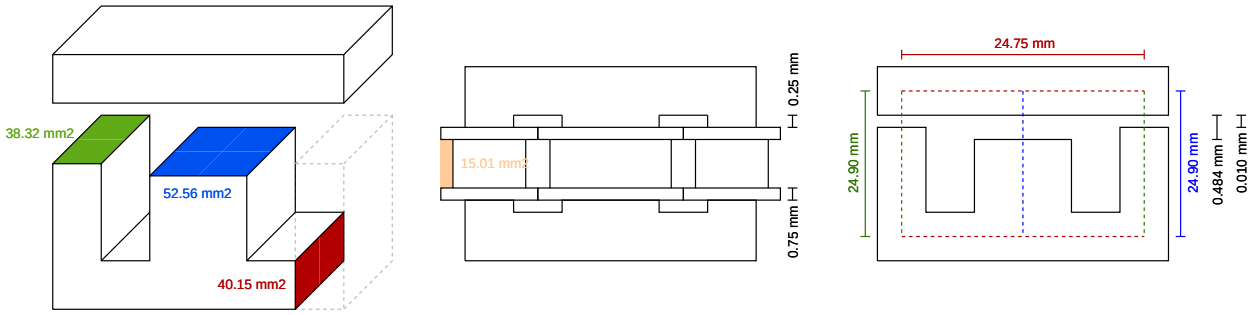
Parameter	Symbol	Value	Unit
Centre-leg gap	lg_center	0.484	mm
Outer-leg gap	lg_outer	0.01	mm

Device drawings

Symbol reference



Calculated values



Note on gap conventions: *The outer-leg gap (lg_outer) is additive. It represents a non-ideal separation between the two core halves (e.g. surface roughness, assembly tolerance) and does not affect the outer-leg magnetic path length. The centre-leg gap (lg_center) is subtractive. It corresponds to material physically removed from the centre leg, reducing its effective magnetic path length. The values shown in the figure are nominal values, calculated based on the core geometry. For the reluctance and inductance calculations, lg_center whill be deducted from le_center*

Main Winding (Centre Leg)

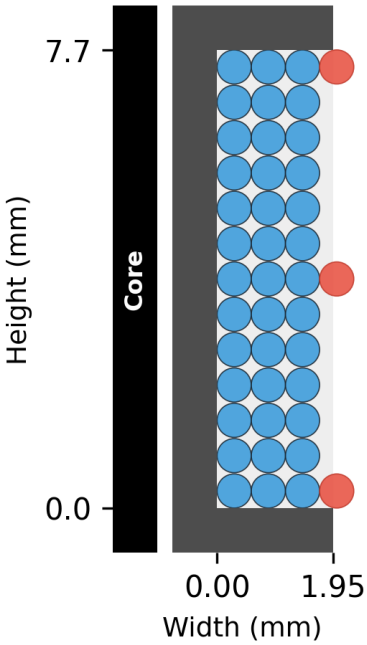
Parameter	Symbol	Value	Unit
Calculated turns	N_main_calc	11.0894	-
Chosen turns	N_main	21	-
AWG	AWG_main	24	-
Parallel conductors	N_cond_main	2	-
Window fill factor required	kw_req_main	72.34	%
Mean turn length	lt_center	40.90	mm
Total wire length	l_main	1.7178	m
Winding resistance	R_main	36.0847	mOhm

Bias Winding (Outer Legs)

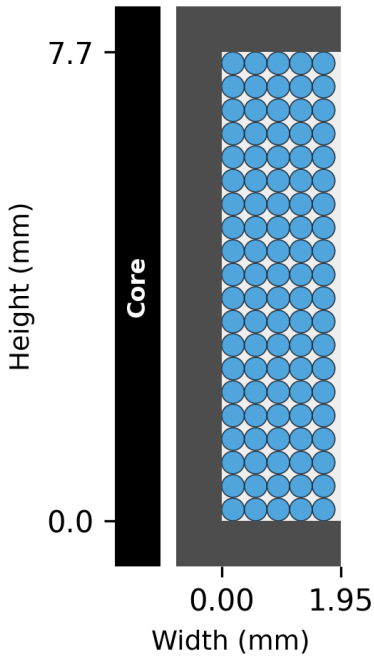
Parameter	Symbol	Value	Unit
Total turns	N_bias	200	-
Turns per outer leg	N_bias/2	100	-
AWG	AWG_bias	28	-
Parallel conductors	N_cond_bias	1	-
Window fill factor required (outer)	kw_bias	72.13	%
Mean turn length (outer)	lt_outer	37.00	mm
Wire length (per leg)	l_bias/leg	3.7000	m
Wire length (total)	l_bias	7.4000	m
Winding resistance (per leg)	R_bias/leg	785.6790	mOhm
Winding resistance (total)	R_bias	1571.3580	mOhm
mu_r needed @ I_bias_max	mu_r_req	7.6	-

Winding cross sections

Main winding - 21T (2 x AWG24)
4 layers - k_w = 72.3%



Bias winding (per leg) - 100T (1 x AWG28)
5 layers - k_w = 72.1%



Operating Points

Parameter	Symbol	Value	Unit
Main winding inductance (I_bias=0)	L_main_nom	60.00	uH
Main winding inductance (I_bias_max)	L_main_min	7.37	uH
MMF main winding (I_main_pk)	F_main	71.4	A·t
Flux centre leg (I_main_pk)	Phi_main	7.4098	uWb
Flux density, centre leg (I_main_pk, I_bias_max)	B_center	140.98	mT
Flux density, outer leg (I_bias_max, linear)	B_outer	252.64	mT
DC flux density, outer leg (I_bias_max, nonlinear)	Bdc_max	505.18	mT

Reluctance Network at I_bias=0 (A·t/Wb)

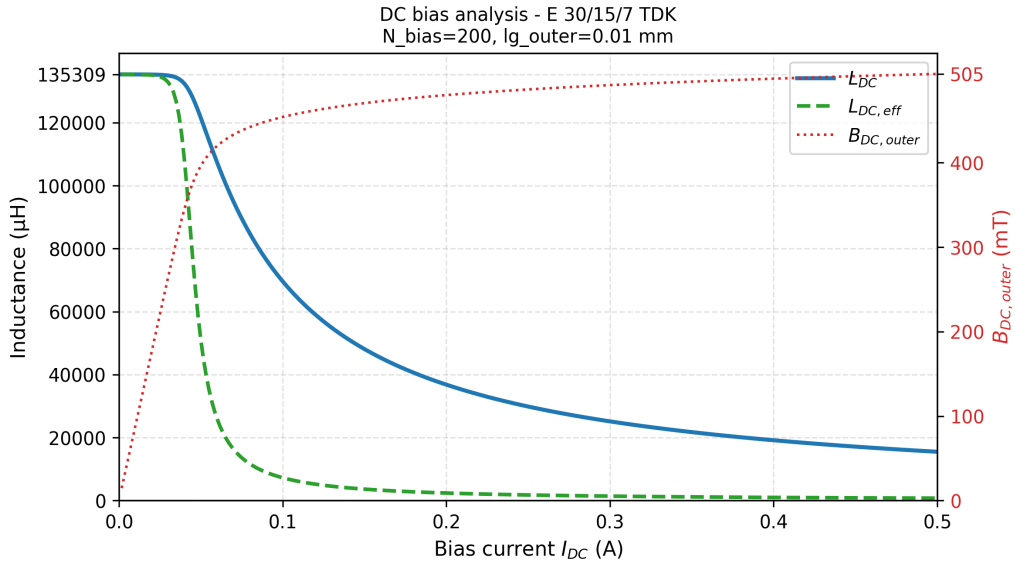
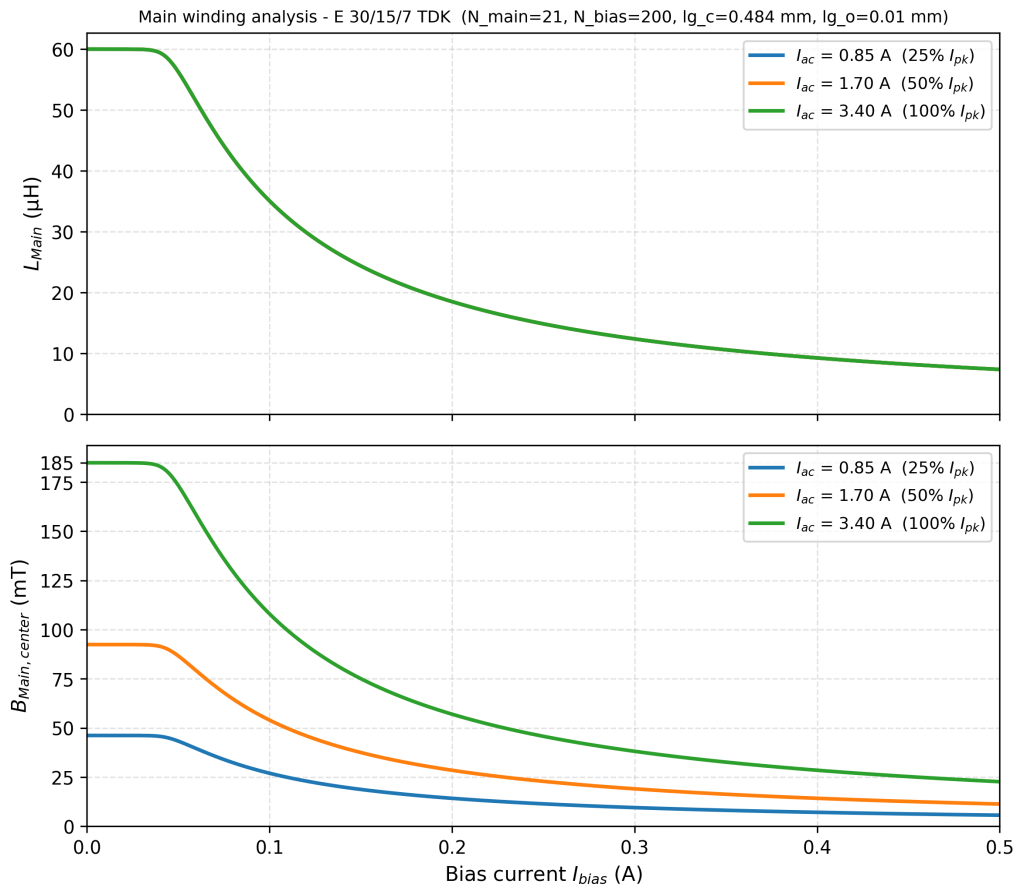
Reluctance element	Symbol	Value	Unit
Horizontal (mu_i)	Rc_h_0	1.8676e+05	A·t/Wb
Outer core (mu_i)	Rc_outer_0	1.9676e+05	A·t/Wb
Outer gap	Rg_outer	2.0764e+05	A·t/Wb
Centre core (linear)	Rc_center	1.4074e+05	A·t/Wb
Centre gap	Rg_center	6.9131e+06	A·t/Wb
Equivalent outer branch	Req_outer_0	5.9116e+05	A·t/Wb
Equivalent centre branch	Req_center	7.0539e+06	A·t/Wb
Total equivalent	Req_total_0	7.3494e+06	A·t/Wb

Reluctance Network at I_bias_max (A·t/Wb)

Reluctance element	Symbol	Value	Unit
Horizontal (nonlinear)	Rc_h	2.4135e+06	A·t/Wb
Outer core (nonlinear)	Rc_outer	2.5428e+06	A·t/Wb
Outer gap	Rg_outer	2.0764e+05	A·t/Wb
Centre core (linear)	Rc_center	1.4074e+05	A·t/Wb
Centre gap	Rg_center	6.9131e+06	A·t/Wb
Equivalent outer branch	Req_outer	5.1640e+06	A·t/Wb
Equivalent centre branch	Req_center	7.0539e+06	A·t/Wb
Total equivalent	Req_total	9.6358e+06	A·t/Wb

Plots

DC inductance and effective DC inductance vs bias current. Solved nonlinearly via mu_T(B).



AC inductance and flux density in the centre leg vs bias current, for several AC current amplitudes. Centre-leg permeability uses mu_D(B_AC).